

**UKFN Special Interest Group in Nuclear Thermal Hydraulics (SIG-NTH) and
Collaborative Computational Project for Nuclear Thermal Hydraulics (CCP-NTH)**

Annual Technical Meeting

29th and 30th June 2026

Whitworth Art Gallery, Oxford Rd, Manchester M15 6ER

Monday 29th June

10:30 - 11:00 Registration & Refreshments

11:00 - 11:15 Welcome Remarks– *Alex? Hector?*

11:15 - 12:00 The UK NTH Community Today (chaired by Alex?)
Round the room ~ 1 minute introduction from each attendee

12:00 - 12:30 **Invited Talk** – TBC – Sanjeev Gupta, Framatome

12:30 - 13:30 Lunch

Session 1 Community and Strategy (chaired by X)

13:30 - 13:50 CCP-NTH Overview - Shuisheng He, The University of Sheffield

13:50 - 14:10 CCP-NTH Research software development and maintenances - Wei Wang, STFC

Session 2 National Nuclear Strategy, Funding and Perspectives (chaired by X)

14:10 - 14:30 **Title(TBC)** - Daniel Mathers, Executive Director, Nuclear Innovation and Research Office (NIRO)

14:30 - 14:50 **Title(TBD)** - Afia Masood, Portfolio Manager, EPSRC, UKRI

14:50 - 14:55 Current Research Topics (*flash talk*) - Sam Mason, Government

Session 3 Experimental Facilities and Safety(chaired by X)

14:55 - 15:15 Experimental investigation of passive natural circulation cooling for Generation IV nuclear reactors - Mohammad Ferdoosi, The University of Manchester

15:15 - 15:35 Updates from the BULLET and THOR facilities at Bangor University - Walter Villanueva, Bangor University

15:35 - 15:55 Progress in higher-fidelity experimental measurements of flow boiling at RR - Brodie Upton, Rolls-Royce Submarines

15:55 - 16:00 Hydrogen Safety in Nuclear Powerplant (*flash talk*) - Yang Zhang, The University of Sheffield

16:00 - 16:30 Coffee Break & Group Photo (for those who gave photo permission)

Session 4 Thermal-Hydraulics Modelling and Industrial Applications (chaired by X)

16:30 - 16:50 Development of numerical modelling approaches for accurate prediction of Natural Circulation Loops - Reuben Petty, The University of Manchester

16:50 - 17:10 Developing a two-phase flow coarse grid model for nuclear reactor thermal-hydraulics – Jeggathishwaran, The University of Sheffield

17:10 - 17:30 System TH Modelling – Challenges and Opportunities in Industry - Christopher Ince, Rolls-Royce PLC

17:30 - 17:35 Transient Analysis of a High-Temperature Gas-Cooled Micro-Reactor for Fission Surface Power Applications using Simulink and NPSS (*flash talk*) - David Barton, Rolls-Royce plc.

17:35 - 17:40 Validation of CFD Modelling for Natural Circulation Loops (*flash talk*) - Thomas John, Rolls-Royce PLC

18:30?

Dinner

Venue...



Tuesday 30th June

08:30 - 09:00

Arrival tea & coffee

Session 5 Data-Driven, Reduced-Order and Multiscale Modelling (chaired by X)

- 09:00 - 09:20 Data-driven boiling predictions: introducing the neural-gated mixture of engineering models approach - Omid Bidar, The University of Sheffield
- 09:20 - 09:40 Reduced-Order Model with Geometry-Conforming, Voxel-Dominated Meshes for Neutron Transport - Andrew Buchan, Queen Mary University of London
- 09:40 - 10:00 **TBD** - Jakov Popov, Queen Mary University of London
- 10:00 - 10:20 Application of Multiscale Multifidelity Methods for Nucleate Boiling in the Nuclear Industry - Matthew Val Meyers, The University of Manchester
- 10:20 - 10:25 **TBD (Talk)** — Patricia Silva, University of Manchester

10:25 - 10:50

Coffee Break

Session 6 Fundamental Physics and HPC (chaired by X)

- 10:50 - 11:20 **Invited talk:** Simulating Tokamak Plasmas for Magnetic Fusion: A CFD-Based Modeling Approach - Eric Serre, CNRS M2P2
- 11:20 - 11:40 The production of plasma turbulence via magnetic reconnection - Nick Williams, The University of Manchester
- 11:40 - 12:00 GPU porting of the DNS solver CHAPSim2 – Bo Liu, STFC UKRI

12:00 – 13:00

Lunch

Session 7 Panel Discussion: Gen-IV Reactor Thermal Hydraulics and Associated Challenges (Chaired by X)

- 13:00 - 13:20 **Invited Talk** – The NTH Nuclear Innovation Programme outcomes and Gen-IV UK landscape – Richard Underhill, Frazer-Nash Consultancy
- 13:20 - 14:20 **Panel Discussion: Panellists:** Chris Sigournay (EDF), Richard Underhill (FNC), Shuisheng He (University of Sheffield), Steve Graham (UKNNL), **Mike Bluck (ICL)**, **Sanjeev Gupta (TBC, Framatome)**
- 14:20 – 14:30** **Closing Remarks – Shuisheng He**

14:30

Departure

Conference Notes

- **Enquiries:** Wei Wang (wei.wang@stfc.ac.uk); Alex Skillen (alex.skillen@manchester.ac.uk)
- **Speakers:** Most speaking slots are 20 minutes. Please aim to present for **15 minutes**, allowing **5 minutes** for audience questions and a smooth transition to the next speaker.
- **Accommodation:** Accommodation for academic attendees and CCP-NTH Advisory Committee members requiring a stay on the night of 29 June 2026 has been arranged and pre-paid at: ****XXXXXX****. Please check in under your own name. Breakfast is included.
- **Conference Dinner**
 - **Venue: **XXXXXX****
 - **Alcohol Policy Notice:** In accordance with UKRI funding guidelines, STFC/EPSCRC funding associated with CCP-NTH cannot be used to purchase alcoholic beverages. Alcohol will therefore not be provided as part of the conference dinner. Attendees are, of course, welcome to purchase their own drinks if they wish. Thank you for your understanding.

